

REAL QUADRATIC PLANES THAT NEED FOUR COLOURS

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ABSTRACT. For a subfield $F \subset \mathbb{R}$ let $\chi(F^2)$ be the chromatic number of the graph on F^2 joining points at Euclidean distance 1. For real quadratic fields $K = \mathbb{Q}(\sqrt{d})$ the values known so far were 2 and 3, and a field can need four colours only when $d \equiv 11 \pmod{12}$. We show that $\chi(\mathbb{Q}(\sqrt{d})^2) = 4$ for $d = 11, 23, 35, 59, 71, 95, 119, 131, 155, 179, 191, 239, 251, 359, 431$, and that $4 \leq \chi(\mathbb{Q}(\sqrt{47})^2) \leq 5$. With the known results, $\chi(\mathbb{Q}(\sqrt{d})^2)$ is now determined for every squarefree $d < 83$ except $d = 47$. Each lower bound is a triangle-free, vertex-critical unit-distance graph with coordinates in $\mathbb{Q}(\sqrt{d})$, with between 94 and 1 404 vertices, that has no proper 3-colouring: a computer proof, by a SAT solver whose DRAT proofs are checked by `drat-trim`, for two independently written encodings. The upper bounds are reductions modulo a prime due to Moorhouse and Fischer. We also explain why some searches fail: when every direction is integral at certain primes above 2, 3 or 5, reduction there colours every graph built from those directions with two or three colours.

1. INTRODUCTION

The Hadwiger–Nelson problem asks for the chromatic number $\chi(\mathbb{R}^2)$ of the plane; since de Grey [3] it is known to be 5, 6 or 7. A natural relative restricts the coordinates to a subfield $F \subset \mathbb{R}$: $\chi(F^2)$ is the least number of colours for the points of F^2 such that points at distance 1 get different colours. Woodall [13] showed $\chi(\mathbb{Q}^2) = 2$. For a real quadratic field $K = \mathbb{Q}(\sqrt{d})$, $d \geq 2$ squarefree, the following was known.

- $\chi(K^2) = 2$ if $d \equiv 1, 2 \pmod{4}$ (Johnson [6], see [11]; Moorhouse [9, Lemma 8.4, Theorem 8.5]).
- $\chi(K^2) \geq 3$ if $d \equiv 3 \pmod{4}$ (Fischer [4, Theorem 8]; Moorhouse [9, Theorem 8.6] for prime d , by an odd cycle from Pell’s equation).
- $\chi(K^2) \leq 3$ if $d \equiv 0, 1 \pmod{3}$ (Fischer [4, Theorem 9]; Moorhouse [9, Corollary 8.3]).
- $\chi(K^2) \leq 4$ if $d \equiv 0, 1, 2, 4 \pmod{7}$ (Moorhouse [9, Corollary 8.3]) or $d \equiv 3 \pmod{8}$ (Fischer [4, Theorem 10]; also [8, Corollary D]). Moorhouse’s Theorem 8.1 states $\chi(K^2) \leq 4$ unless $d \equiv 47, 59, 83 \pmod{84}$.

Madore [7] proved $\chi(\mathbb{Q}(\sqrt{3})^2) = \chi(\mathbb{Q}(\sqrt{7})^2) = 3$. Together these give $\chi(K^2) = 3$ when $d \equiv 3 \pmod{4}$ and $d \not\equiv 2 \pmod{3}$; so a real quadratic field can need four colours only if $d \equiv 11 \pmod{12}$, and for those the known lower bound was 3. As far as we could find, no real quadratic field was known to need four colours. Moorhouse asked in a 2010 talk [10] what can be said about $\chi(\mathbb{Q}(\sqrt{d})^2)$, and about $\chi(\mathbb{Q}(\sqrt{47})^2)$ in particular.

Theorem 1. $\chi(\mathbb{Q}(\sqrt{d})^2) = 4$ for $d = 11, 23, 35, 59, 71, 95, 119, 131, 155, 179, 191, 239, 251, 359, 431$, and $4 \leq \chi(\mathbb{Q}(\sqrt{47})^2) \leq 5$.

Corollary 2. For squarefree $2 \leq d < 83$, $\chi(\mathbb{Q}(\sqrt{d})^2)$ is 2 if $d \equiv 1, 2 \pmod{4}$, 3 if $d \equiv 3 \pmod{4}$ and $d \not\equiv 2 \pmod{3}$, and 4 if $d \in \{11, 23, 35, 59, 71\}$. The only remaining value is $d = 47$, with $4 \leq \chi \leq 5$.

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The upper bounds in Theorem 1 are known (Section 4). Each lower bound is a finite graph: a unit-distance graph with coordinates in $\mathbb{Q}(\sqrt{d})$ that is not 3-colourable. That it is not 3-colourable is a computer proof (Section 3); the rest of each certificate is checked in exact integer arithmetic. The graphs, the formulas, the solver logs, the checker and the search code are in the repository [5].

The graphs are triangle-free: a unit triangle needs $\sqrt{-3} \in K(i)$, which fails for $d \neq 3$. A triangle-free 5-chromatic unit-distance graph is not known (see [12, ch. 55], [8, §2]); since $\chi(\mathbb{Q}(\sqrt{47})^2) \leq 5$ and $\sqrt{3} \notin \mathbb{Q}(\sqrt{47})$, deciding whether $\chi(\mathbb{Q}(\sqrt{47})^2)$ is 4 or 5 touches that question.

2. THE GRAPHS

Fix d and a positive integer D . A point is written $[a, b, c, e]$ with $a, b, c, e \in \mathbb{Z}$, and stands for

$$\left(\frac{a + b\sqrt{d}}{D}, \frac{c + e\sqrt{d}}{D} \right) \in K^2.$$

Since 1 and $\sqrt{d}$ are linearly independent over $\mathbb{Q}$, two points are at distance 1 exactly when their differences satisfy the two integer identities

$$(1) \quad \Delta a^2 + d \Delta b^2 + \Delta c^2 + d \Delta e^2 = D^2, \quad \Delta a \Delta b + \Delta c \Delta e = 0.$$

So every edge is checked exactly. Table 1 lists the graphs, and Figure 1 shows the one over $\mathbb{Q}(\sqrt{11})$.

d	D	vertices	edges	degrees	directions used
11	30	94	214	3–32	68
23	120	660	1 727	3–60	100
35	390	580	1 501	3–107	218
47	240	816	2 134	3–72	94
59	210	406	993	3–48	66
71	120	611	1 557	3–85	128
95	480	1 404	3 780	3–99	126
119	240	399	1 019	3–62	106
131	390	356	804	3–61	126
155	510	1 281	3 526	3–103	124
179	390	259	622	3–39	90
191	240	100	224	3–23	52
239	480	355	888	3–47	80
251	390	291	715	3–56	98
359	600	715	1 851	3–63	74
431	600	331	764	3–28	66

TABLE 1. The graphs of Theorem 1. D is the common denominator; “directions used” counts the unit vectors $v - u$ over the edges uv , with both signs.

For each graph G :

- the edges are all the pairs of its points that satisfy (1);
- there is no triangle, and the girth is 4;
- there is a proper 4-colouring, and no proper 3-colouring;
- G is vertex-critical: for every vertex v there is a proper 3-colouring of $G - v$.

3. THE CERTIFICATES

For a graph G with vertices $0, \dots, n-1$ and a fixed edge $u_0 v_0$, let $\Phi(G)$ be the formula with variables $x_{v,c}$ (v a vertex, $c \in \{0, 1, 2\}$), the clauses $x_{v,0} \vee x_{v,1} \vee x_{v,2}$ for each vertex, $\neg x_{u,c} \vee \neg x_{v,c}$ for each edge uv and colour c , and the unit clauses $x_{u_0,0}$ and $x_{v_0,1}$. Renaming colours, G is 3-colourable if and only if $\Phi(G)$ is satisfiable.

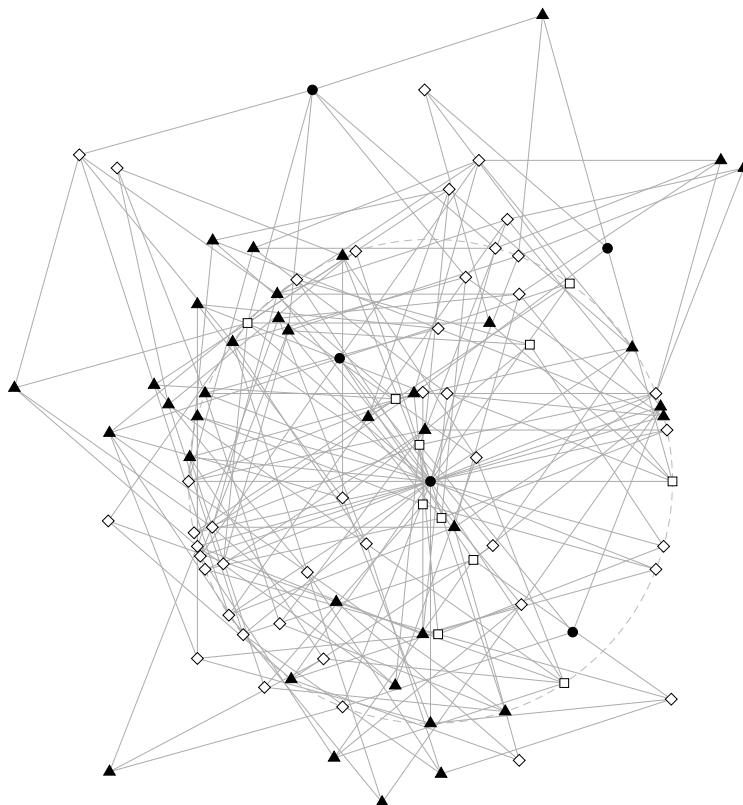


FIGURE 1. The graph over $\mathbb{Q}(\sqrt{11})$: 94 points, joined by the 214 segments of length exactly 1 (drawn from the coordinates rounded to five decimals; `make_figure.py` writes the drawing from the data). The marker shapes show a proper 4-colouring; there is no proper 3-colouring. The vertex at the origin has 32 neighbours, on the dashed unit circle.

For each d the repository stores $\Phi(G)$, with $x_{v,c}$ numbered $3v + c + 1$. The solver kissat 4.0.4 answered UNSATISFIABLE and wrote a DRAT proof, which `drat-trim` verified. The same was done for a second encoding, written by separate code, with $x_{v,c}$ numbered $cn + v + 1$. The DRAT proofs are a few megabytes and each check takes seconds.

The checker `scripts/verify_quadratic_planes.py`, which uses only the Python standard library, checks again, for each d : that the points are distinct, that every listed edge satisfies (1) and every pair satisfying (1) is listed, that there is no triangle, the stored 4-colouring, the stored 3-colourings of $G - v$ for every v , that the stored formula is exactly $\Phi(G)$, and the colourings of the finite planes behind the upper bounds. With a path to kissat and `drat-trim` it also writes the second encoding, solves it and checks the proof. With a path to `cake_lpr` as well, `drat-trim` writes each proof in LRAT form and `cake_lpr`, a proof checker verified in HOL4 and compiled by the verified CakeML compiler [1], checks it; it accepted the proofs for all sixteen graphs. What has to be trusted is a proof checker (`drat-trim`, or `cake_lpr` and its formal verification), the short code that writes $\Phi(G)$ from the list of edges, and exact integer arithmetic.

4. THE UPPER BOUNDS

Lemma 3 (Moorhouse [9, Lemma 8.2]). *If $p \equiv 3 \pmod{4}$ is prime and d is 0 or a nonzero square modulo p , then $\chi(\mathbb{Q}(\sqrt{d})^2) \leq \chi(\mathbb{F}_p^2)$, where $\mathbb{F}_p^2$ carries the graph $x^2 + y^2 = 1$.*

The prime p ramifies or splits in K ; at a prime $\mathfrak{p}$ of norm p the form $x^2 + y^2$ is anisotropic over $\mathbb{F}_p$, so every unit vector is $\mathfrak{p}$ -integral and reduction modulo $\mathfrak{p}$ is a graph homomorphism on the component of the origin. Since $\chi(\mathbb{F}_7^2) = 4$ [9, §7] and $\mathbb{F}_{11}^2$ has a proper 5-colouring [7, Lemma 4.5]:

- $d = 11, 23, 71, 95, 155, 179, 191, 239, 359, 431$ are nonzero squares and $d = 35, 119$ are $\equiv 0$ modulo 7, so $\chi \leq 4$;
- $d = 59, 131, 251$ are $\equiv 3 \pmod{8}$, so $\chi \leq 4$ by Fischer's Theorem 10 (which also covers 11, 35 and 179);
- $47 \equiv 5^2 \pmod{11}$, so $\chi(\mathbb{Q}(\sqrt{47})^2) \leq 5$.

The repository stores a proper 4-colouring of $\mathbb{F}_7^2$ and a proper 5-colouring of $\mathbb{F}_{11}^2$, and the checker checks them. For $d = 47$ the next level at 11 does not help: the unit-distance graph on $(\mathbb{Z}/121)^2$ is not 4-colourable.

5. HOW THE GRAPHS WERE FOUND

Directions. For a denominator D let U_D be the set of unit vectors $[a, b, c, e]$ with integer coordinates, that is, of solutions of (1) with $\Delta = (a, b, c, e)$. It is finite, and closed under negation, the quarter turn and complex conjugation. We used sets with 108 to 276 elements.

Colouring-guided growth. Start from all sums of at most two vectors of U_D ; 3-colour the graph (tabu search, else kissat); add the candidate points $p + u$ (p in the graph, $u \in U_D$) whose neighbours in the graph already see all three colours; repeat until kissat answers that the graph is not 3-colourable. When no candidate was blocked, a variant coloured the graph again from scratch and, failing that, added candidates whose neighbours see two colours. The successful runs took 1 to 96 rounds and ended with 5 941 to 39 265 points.

Shrinking. A single incremental SAT solver (CaDiCaL through PySAT) holds the formula with one selector literal per vertex, enabling its at-least-one clause. Each test is a solve under assumptions, and each refutation returns a core of selectors, a subset of the vertices that is still not 3-colourable. Deleting vertices one at a time, keeping a deletion when the rest is still refuted, gives a vertex-critical graph; repeating with up to 400 random deletion orders and keeping the smallest result gave the graphs of Table 1.

Gates. Some searches stop with 3-colourable graphs for an arithmetic reason. Let $\mathfrak{P}$ be a prime of $K(i)$ above p that splits in $K(i)/K$, with residue field $\mathbb{F}_q$. Writing a unit vector as $z = x + iy$, we have $z\bar{z} = 1$, where the bar is the automorphism $i \mapsto -i$ over K ; so $v_{\mathfrak{P}}(z) = -v_{\bar{\mathfrak{P}}}(z)$, and if z is $\mathfrak{P}$ -integral its residues at $\mathfrak{P}$ and $\bar{\mathfrak{P}}$ are inverse to each other. Hence, if every direction is $\mathfrak{P}$ -integral, the reduction $z \mapsto (z \bmod \mathfrak{P}, z \bmod \bar{\mathfrak{P}})$ maps every graph built from the directions homomorphically into

$$H_q = \text{Cay}(\mathbb{F}_q^2, \{(x, x^{-1}) : x \in \mathbb{F}_q^\times\}),$$

and $\chi(H_q) = 3$ for $q = 3, 5, 9$, while $\chi(H_q) \geq 5$ for $q = 13, 17, 25, 29, 37, 41, 49$ (computed with kissat). The place above 2 splits with residue field $\mathbb{F}_2$ when $d \equiv 7 \pmod{8}$, and there the graph is a perfect matching. This gives three *gates*:

- at 2, when $d \equiv 7 \pmod{8}$ (two colours): with the 44 vectors $s/\bar{s}$ for s with coefficients in $\{-1, 0, 1\}$, every graph over $\mathbb{Q}(\sqrt{47})$ we built was bipartite;
- at 3, when $d \equiv 2 \pmod{3}$, with residue field $\mathbb{F}_9$: for $d = 83$ and $D = 410$ every direction is integral there;
- at 5, when $d \equiv 0, \pm 1 \pmod{5}$, with residue field $\mathbb{F}_5$: for $d = 35$ and $D = 174$ every direction is integral there.

Each of these direction sets builds only 3-colourable graphs. A useful set contains vectors that are integral at no gate, such as $(1 + 7i\sqrt{47})/48$, from Pell's equation $48^2 - 47 \cdot 7^2 = 1$. The gates do not explain every failure: the first searches over $\mathbb{Q}(\sqrt{35})$, $\mathbb{Q}(\sqrt{131})$ and $\mathbb{Q}(\sqrt{179})$ failed with sets of 60 to 212 directions that pass all three gates, and succeeded with the 196 to 276 directions of denominator dividing 390 and the variant of the growth.

6. REMARKS

A different question: $\mathbb{Q}(\sqrt{-d})$ inside $\mathbb{C}$. Cohen [2] wrote that “the natural conjecture is that $\mathbb{Q}[\alpha]$ is 3-colorable for all α quadratic over $\mathbb{Q}$ ”, for the set $\mathbb{Q}[\alpha] \subset \mathbb{C}$. For $\alpha = \sqrt{-d}$ this set is $\{x + y\sqrt{-d}\}$, of dimension 2 over $\mathbb{Q}$, while $\mathbb{Q}(\sqrt{d})^2$ has dimension 4; so Theorem 1 does not contradict the conjecture. The conjecture holds:

Proposition 4. *For every α quadratic over $\mathbb{Q}$, the unit-distance graph on $\mathbb{Q}[\alpha] \subset \mathbb{C}$ is 3-colourable.*

Proof. For real α the set lies on a line. For $\alpha = \sqrt{-1}$ it is $\mathbb{Q}^2$ [13], and for $\alpha = \sqrt{-2}$ it lies in $\mathbb{Q}(\sqrt{2})^2$, which is 2-colourable. Otherwise d has an odd prime factor p , and the prime $\mathfrak{p}$ of $L = \mathbb{Q}(\sqrt{-d})$ above p is ramified. An element $z \in L$ with $|z| = 1$ satisfies $z\bar{z} = 1$ and $v_{\mathfrak{p}}(\bar{z}) = v_{\mathfrak{p}}(z)$, so z is a $\mathfrak{p}$ -adic unit; and $z \equiv \bar{z} \pmod{\mathfrak{p}}$, so $z^2 \equiv 1$ and $z \equiv \pm 1 \pmod{\mathfrak{p}}$. Choose a representative r_C of each coset C of the ring $\mathcal{O}_{\mathfrak{p}}$ of $\mathfrak{p}$ -integral elements, and colour $w \in C$ by $f((w - r_C) \bmod \mathfrak{p})$, where f is a proper 3-colouring of the cycle $\text{Cay}(\mathbb{F}_p, \{\pm 1\})$. Points at distance 1 differ by such an element, lie in the same coset, and have residues differing by ± 1 . $\square$

Open questions.

- Is $\chi(\mathbb{Q}(\sqrt{47})^2)$ equal to 4 or 5? A value of 5 would give a triangle-free 5-chromatic unit-distance graph.
- Does every real quadratic field with $d \equiv 11 \pmod{12}$ need four colours? Our searches over $\mathbb{Q}(\sqrt{83})$, $\mathbb{Q}(\sqrt{107})$ and $\mathbb{Q}(\sqrt{203})$ have so far found only 3-colourable graphs.
- A uniform proof: a family of graphs, or an argument, covering all $d \equiv 11 \pmod{12}$ at once.
- How small can a 4-chromatic unit-distance graph over a real quadratic field be? Ours has 94 vertices for $d = 11$ and 100 for $d = 191$.

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